Journey Through the Solar System

Key facts about the Sun, the planets and space exploration

The Sun and the Birth of the Solar System

The Solar System formed about 4.6 billion years ago from a giant cloud of gas and dust. As part of the cloud collapsed under its own gravity, most of the material fell into the centre and formed the Sun. The leftover material flattened into a spinning disc, and inside that disc the planets slowly grew.

The Sun is by far the dominant object in the Solar System. It contains about 99.86 percent of the total mass of the whole system. It is a medium-sized star made mostly of hydrogen and helium, and in its core hydrogen is fused into helium, releasing the energy that gives us light and heat.

Light from the Sun takes about eight minutes and twenty seconds to travel to Earth. That means when you look at the Sun, you are seeing it as it was a little over eight minutes ago. The Sun is expected to keep shining in a stable way for roughly another five billion years.

There are eight planets in the Solar System. In order from the Sun they are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. The first four are small, rocky worlds called the terrestrial planets. The last four are much larger. Jupiter and Saturn are gas giants, while Uranus and Neptune are often called ice giants.

Between Mars and Jupiter lies the asteroid belt, a wide region filled with rocky bodies left over from the formation of the planets. The largest object in the belt, Ceres, is classified as a dwarf planet.

The Inner Planets: Mercury, Venus, Earth and Mars

Mercury is the smallest planet and the closest to the Sun. It races around the Sun in just 88 Earth days, which makes its year the shortest of any planet. It has no moons and almost no atmosphere, so temperatures swing wildly between day and night.

Venus is the hottest planet, even though Mercury is closer to the Sun. Its thick atmosphere of carbon dioxide traps heat and keeps the surface at around 465 degrees Celsius. Venus rotates very slowly and in the opposite direction to most planets. One rotation takes about 243 Earth days, which is longer than its year of about 225 Earth days.

Earth is the only planet known to support life. About 71 percent of its surface is covered by liquid water, and it has a single large natural satellite, the Moon. Earth takes 365.25 days to orbit the Sun, which is why we add a leap day to the calendar every four years.

Mars is known as the Red Planet because iron oxide, or rust, covers much of its surface. It has two small moons, Phobos and Deimos. Mars is home to Olympus Mons, the tallest known volcano in the Solar System, which rises about 22 kilometres above the surrounding plain. That is roughly two and a half times the height of Mount Everest above sea level.

Many robotic missions have explored Mars with orbiters, landers and rovers. Scientists study its surface to learn whether liquid water once flowed there and whether the planet might once have been suitable for life.

The Giant Planets: Jupiter and Saturn

Jupiter is the largest planet. It is so big that more than 1,000 Earths could fit inside it. It is made mostly of hydrogen and helium and has no solid surface that a spacecraft could land on. Its most famous feature is the Great Red Spot, a giant storm that has been observed for well over a century and is wider than Earth.

Jupiter has dozens of moons. The four largest, Io, Europa, Ganymede and Callisto, were discovered by Galileo Galilei in 1610. Ganymede is the largest moon in the entire Solar System, and it is even bigger than the planet Mercury. Europa is of special interest because a deep ocean of liquid water probably lies beneath its icy crust.

Saturn is the second-largest planet and is famous for its spectacular rings. The rings are made mostly of countless pieces of water ice, ranging in size from tiny grains to chunks as large as a house. Although the rings are very wide, they are extremely thin, in places only about ten metres thick.

Saturn has an unusually low density. It is less dense than water, which means that in a bathtub big enough to hold it, Saturn would float. Its largest moon, Titan, is the only moon in the Solar System with a thick atmosphere, and it has lakes and seas of liquid methane and ethane on its surface.

Saturn winds can reach more than 1,000 kilometres per hour near the equator. At its north pole there is a strange six-sided cloud pattern, known as the hexagon, which scientists first noticed in images taken by the Voyager spacecraft.

The Ice Giants and Beyond: Uranus, Neptune and Pluto

Uranus was discovered in 1781 by the astronomer William Herschel, who first thought it was a comet. It is tilted over on its side, with its axis of rotation at about 98 degrees, so it seems to roll around the Sun like a ball. Each of its poles gets about 42 years of continuous sunlight followed by 42 years of darkness.

Neptune is the farthest planet from the Sun. It was found in 1846, after mathematicians calculated where an unseen planet must be in order to explain irregularities in the orbit of Uranus. Astronomers at the Berlin Observatory pointed their telescope at the predicted position and saw Neptune on the very same night.

Neptune is a cold, windy world. Its winds are the fastest in the Solar System, reaching about 2,000 kilometres per hour. Because it is so far from the Sun, one Neptunian year lasts about 165 Earth years. Neptune completed its first full orbit since its discovery in 2011.

Pluto was discovered in 1930 and was called the ninth planet for 76 years. In 2006 the International Astronomical Union created a formal definition of a planet. Pluto did not meet all the requirements, because it has not cleared other objects from the neighbourhood of its orbit, and it was reclassified as a dwarf planet.

Beyond Neptune lies the Kuiper Belt, a vast ring of icy bodies that includes Pluto and several other dwarf planets. Much farther out is the Oort Cloud, a theoretical shell of icy objects that may be the source of long-period comets.

Exploring the Solar System

Humans have explored the Solar System with spacecraft for more than sixty years. In July 1969 the Apollo 11 mission landed the first people on the Moon. Neil Armstrong and Buzz Aldrin walked on the surface while Michael Collins orbited above in the command module.

The twin Voyager spacecraft were launched in 1977 to study the outer planets. Voyager 1 passed Jupiter and Saturn, and in 2012 it became the first human-made object to enter interstellar space. Voyager 2 is the only spacecraft that has ever visited Uranus and Neptune.

Robotic explorers have landed on Venus, Mars, Titan and several asteroids and comets. The Cassini spacecraft orbited Saturn from 2004 to 2017, and it carried the Huygens probe that landed on Titan in 2005. The New Horizons spacecraft flew past Pluto in July 2015, sending back the first close-up images of its surface.

Telescopes in space complete the picture. The Hubble Space Telescope, launched in 1990, has produced thousands of detailed images of the planets, and the James Webb Space Telescope, launched in December 2021, observes the sky in infrared light.

Looking ahead, space agencies are planning new missions to return humans to the Moon, to bring rock samples back from Mars, and to explore the icy moons of Jupiter and Saturn in search of signs of life.